Do Prediction Market Prices Equal Probabilities? Evidence from a Play-Money Contrast

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Abstract

Prediction market prices are widely interpreted as event probabilities, yet whether they embed systematic distortions is difficult to assess without either an external benchmark or a credible contrast case. This note exploits a natural contrast: the play-money platform Manifold, where participants face no financial stakes. Using a one-parameter Wang specification as a measurement device, I estimate platform-level price–probability wedges on 290,730 resolved contracts across six platforms. The two traded real-money venues, Polymarket and Kalshi, yield positive and highly significant estimates, while Manifold yields a negative estimate—a sign reversal consistent with confidence-related distortions becoming relatively more important when financial stakes are absent. Three forecasting platforms also exhibit positive estimates, though their economic interpretation differs from traded venues. The cross-platform sign pattern is robust to sample restrictions and complement-invariant specifications. The results provide compact evidence that prediction market prices need not equal physical probabilities and that play-money markets offer a useful contrast case for interpreting price–probability wedges.

JEL Classification: G13, G14, D81

Keywords: prediction markets, Wang transform, price–probability wedge, play-money markets, favorite-longshot bias

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1 Introduction

Prediction market prices are routinely read as event probabilities by media, policymakers, and market participants (Wolfers and Zitzewitz, 2004). Yet Manski (2006) showed that in general, equilibrium prices in markets with heterogeneous beliefs need not equal mean beliefs or physical probabilities. Testing whether prediction market prices embed systematic distortions is difficult because the physical probability of a unique real-world event is fundamentally unobservable.

Prediction markets are nonetheless widely used as probability proxies (Wolfers and Zitzewitz, 2004), and prior work comparing real-money and play-money venues has found broadly similar accuracy (Servan-Schreiber et al., 2004), though systematic distortion patterns have not been examined through a unified measurement framework.

This note sidesteps the benchmark problem by exploiting a *natural contrast*: the play-money platform Manifold, where participants trade tokens with no financial value. If the price–probability wedge on real-money platforms reflects risk-bearing costs—inventory risk, capital lockup, adverse selection (Whelan, 2024; Shin, 1991)—then removing financial stakes should attenuate or possibly reverse the wedge. If instead the wedge mainly reflects cognitive distortions, it should be more likely to persist with similar sign.

I estimate a one-parameter Wang transform specification on 290,730 resolved contracts across six platforms. The main finding is a *sign reversal*: traded real-money venues (Polymarket, Kalshi) exhibit positive wedges, while Manifold exhibits a negative wedge. This pattern is consistent with the interpretation that risk-bearing forces are more important on traded venues, while confidence-related distortions dominate when financial stakes are absent.

This note uses the Wang transform purely as a measurement device. A separate working paper (Yang, 2026) develops one structural foundation for this specification.

2 Measurement framework

The Wang (2000) transform maps physical event probabilities into market prices via a single parameter:

$$p^{\mathrm{mkt}} = \Phi\big(\Phi^{-1}(p^*) + \lambda\big)\,, \tag{1}$$

where Φ is the standard normal CDF, p^* is the physical probability, and λ is the price–probability wedge parameter. When $\lambda > 0$, market prices lie systematically above physical probabilities; when $\lambda < 0$, they lie below.

For a cross-section of N resolved contracts $(p_i^{\mathrm{mkt}}, Y_i)$, where $Y_i \in \{0,1\}$ is the resolution outcome, the maximum-likelihood estimator is:

$$\hat{\lambda}_{\mathrm{MLE}} = \arg\max_{\lambda} \sum_{i=1}^{N} \left[Y_i \ln \Phi\big(\Phi^{-1}(p_i^{\mathrm{mkt}}) - \lambda\big) + (1 - Y_i) \ln\big(1 - \Phi\big(\Phi^{-1}(p_i^{\mathrm{mkt}}) - \lambda\big)\big) \right].$$

Standard errors are from the observed Fisher information. Throughout, λ is used as a one-parameter measurement device for price–probability wedges; this note does not claim to separately identify the structural source of the estimated parameter.

3 Data

I estimate platform-level wedges on six platforms spanning three distinct institutional architectures:

- **Traded real-money venues.** Polymarket (blockchain-based central limit order book (CLOB); N = 14,723, pooling 13,738 contracts from the 2026 CLOB sample and 985 from a 2025 HuggingFace archive) and Kalshi (CFTC-regulated exchange; N = 271,699; 2021–2026). Participants risk real capital.
- **Forecasting tournaments.** Metaculus (N = 1,845), Good Judgment Open (N = 692), and INFER (N = 90), sourced from HuggingFace (2015–2024). Participants submit probability estimates under reputational or prize-based incentives. These platforms are included as ancillary evidence; their economic object differs from traded venues since participants do not bear direct financial risk.
- **Play-money platform.** Manifold (N = 1,681), sourced from HuggingFace (2015–2024). Participants trade with tokens carrying no financial value, providing a natural contrast that removes risk-bearing costs while preserving cognitive factors.

All contracts are filtered to $p \in (0.02, 0.98)$ at the platform level; the reported counts reflect each platform's full filtered sample as used in the cross-platform analysis. Total sample: N = 290,730 resolved contracts across six platforms spanning 2015–2026.[1]

[1] A separate working paper (Yang, 2026) reports a pooled total of 291,309 contracts including 579 Polymarket observations sourced from a multi-platform forecasting dataset. Those observations are excluded here to maintain a clean one-row-per-platform structure.

4 Results

Table 1 reports the central result: the sign of the estimated Wang parameter differs between traded real-money venues and the play-money platform.

Table 1: Platform-level Wang parameter estimates. Traded real-money venues (Panel A) and the play-money platform (Panel C) are the primary comparison; forecasting tournaments (Panel B) provide ancillary evidence. Standard errors from observed Fisher information; significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

Platform	N	$\hat{\lambda}_{\text{MLE}}$	SE	Significance
Panel A: Traded real-money venues				
Polymarket (pooled)	14,723	0.164^{***}	0.011	$p < 10^{-15}$
Kalshi (2021–2026)	271,699	0.187^{***}	0.003	$p < 10^{-15}$
Panel B: Forecasting tournaments (ancillary)				
Metaculus	1,845	0.287^{***}	0.033	$p < 10^{-15}$
Good Judgment Open	692	0.570^{***}	0.055	$p < 10^{-15}$
INFER	90	0.635^{***}	0.180	$p < 0.001$
Panel C: Play-money platform				
Manifold	1,681	-0.218^{***}	0.032	$p < 10^{-11}$

Three findings emerge. First, both traded real-money venues yield positive and highly significant estimates ($\hat{\lambda} \approx 0.16$–$0.19$), indicating a positive price–probability wedge. Second, Manifold yields $\hat{\lambda} = -0.218$—a sign reversal. Removing financial stakes does not merely attenuate the wedge; it flips its direction. This is consistent with confidence-related distortions (overconfidence, miscalibration) dominating when risk-bearing costs are absent. Third, forecasting tournaments also exhibit positive estimates, though typically larger in magnitude. The economic interpretation of λ on these platforms differs from traded venues: participants submit probability assessments rather than placing capital at risk, so the positive distortion may reflect different mechanisms (e.g., extremity aversion in forecast elicitation rather than inventory risk).

The magnitude gradient across traded venues—Polymarket ($\hat{\lambda} \approx 0.17$) slightly below Kalshi ($\hat{\lambda} \approx 0.19$)—is consistent with the view that deeper, more liquid markets compress the wedge further, though differences in price timing and contract ecology preclude a clean causal interpretation.

5 Robustness

The sign reversal is robust along several dimensions. Re-estimating with alternative price measures (5-price opening average, midlife price, near-settlement price) preserves the positive sign on Polymarket and Kalshi and the negative sign on Manifold. Using a complement-invariant specification that reorients each contract to the less-likely outcome attenuates the magnitude on all platforms but preserves the sign pattern. Restricting to $p \in (0.10, 0.90)$ or to contracts with duration > 7 days leaves the sign pattern unchanged.

6 Conclusion

Prediction market prices on traded real-money venues embed a positive price–probability wedge that reverses sign on the play-money platform Manifold, consistent with risk-bearing costs dominating on traded venues and confidence-related distortions dominating absent financial stakes. Identifying the structural source of the wedge requires richer microdata or exogenous variation; but the sign reversal itself shows that prediction market prices cannot always be read literally as probabilities.

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